We can now see how to determine the valuation v corresponding to a discrete valuation ring A. Given a discrete valuation ring A with unique maximal ideal $\mathfrak{m}$, we may define $v\colon A \to \mathbb{Z}$ by letting $v(a)$ be the unique integer n for which $(a) = \mathfrak{m}^n$ and $v(0) := \infty$. Extending v to the fraction field k of A via $v(a/b) := v(a) - v(b)$ gives a discrete valuation v on k for which $A = \{x \in k : v(x) \geq 0\}$ is the corresponding valuation ring.

Notice that any discrete valuation v on k with A as its valuation ring must satisfy $v(\pi) = 1$ for some $\pi \in \mathfrak{m}$ (otherwise $v(k) \neq \mathbb{Z}$), and we then have $v(\pi) = 1$ if and only if $\mathfrak{m} = (\pi)$. Moreover, v must then coincide with the discrete valuation we just defined: for any DVR A, the discrete valuation on the fraction field of A that yields A as its valuation ring is uniquely determined. It follows that we could have defined a uniformizer to be any generator of the maximal ideal of A without reference to a valuation.

Example 1.14. For the p-adic valuation $v_p\colon \mathbb{Q} \to \mathbb{Z} \cup \{\infty\}$ we have the valuation ring

$$\mathbb{Z}_{(p)} := \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, p \nmid b \right\},$$

with maximal ideal $\mathfrak{m} = (p)$; this is the *localization* of the ring $\mathbb{Z}$ at the prime ideal (p). The residue field is $\mathbb{Z}_{(p)}/p\mathbb{Z}_{(p)} \simeq \mathbb{Z}/p\mathbb{Z} \simeq \mathbb{F}_p$.

Example 1.15. For any field k, the valuation $v\colon k((t)) \to \mathbb{Z} \cup \{\infty\}$ on the field of Laurent series over k defined by

$$v\left(\sum_{n \geq n_0} a_n t^n\right) = n_0,$$

where $a_{n_0} \neq 0$, has valuation ring $k[[t]]$, the power series ring over k. For $f \in k((t))^\times$, the valuation $v(f) \in \mathbb{Z}$ is the *order of vanishing* of f at zero. For every $\alpha \in k$ one can similarly define a valuation v_α on k as the order of vanishing of f at α by taking the Laurent series expansion of f about α.

1.5 Discrete Valuation Rings

Discrete valuation rings are in many respects the nicest rings that are not fields. In addition to being an integral domain, every discrete valuation ring A enjoys the following properties:

- *noetherian*: Every increasing sequence $I_1 \subseteq I_2 \subseteq \cdots$ of ideals eventually stabilizes; equivalently, every ideal is finitely generated.
- *principal ideal domain*: Every ideal is principal (generated by a single element).
- *local*: There is a unique maximal ideal $\mathfrak{m}$.
- *dimension one*: The (Krull) *dimension* of a ring R is the supremum of the lengths n of all chains of prime ideals $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$ (which need not be finite, in general). For DVRs, $(0) \subseteq \mathfrak{m}$ is the longest chain of prime ideals, with length 1.
- *regular*: The dimension of the $A/\mathfrak{m}$-vector space $\mathfrak{m}/\mathfrak{m}^2$ is equal to the dimension of A. Non-local rings are regular if this holds for every localization at a prime ideal.
- *integrally closed* (or *normal*): Every element of the fraction field of A that is the root of a monic polynomial in $A[x]$ lies in A.
- *maximal*: There are no intermediate rings strictly between A and its fraction field.